

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

(35)

**COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO5:

Measure network performance using key metrics with simulation tools
(BL4, BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements 1	Identifies most constraints with minor gaps in understanding 1	Identifies some constraints but misses key aspects 1	Shows little or no understanding of constraints or problem 2
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively 1	Breaks problem into sub-parts with some logical flow 1	Attempts decomposition but lacks clarity or structure 1	No clear decomposition of the problem 2
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design 1	Applies most constraints correctly with minor errors 1	Applies some constraints but with noticeable errors 1	Fails to apply constraints appropriately 1
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies 1	Uses generally sound reasoning with minor gaps 1	Reasoning is partially correct but inconsistent 1	Reasoning is unclear or incorrect 1
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized 1	Solution is mostly correct and feasible 1	Solution has errors or is partially feasible 1	Solution is incorrect or not feasible 1
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach 2	Shows some creativity in approach 1	Limited creativity shown 1	No creativity; relies on basic or copied ideas 2
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation 2	Presentation is clear with minor issues 2	Presentation lacks clarity or organization 2	Poorly presented and difficult to understand 1

9

8

6

10

for a Remote Rural Branch.

for a Remote rural Branch where wired high speed Internet is unavailable; a T1/56kbps connection or satellite Internet is suitable.

Primary Connection: T1/56kbps fixed wireless Internet, if cellular coverage is available.

Backup Connection: Satellite Internet or a second cellular provider.

use dual-WAN / fail over so that if the primary connection fails, traffic automatically switches to the backup.

A business grade connection with a suitable data plan should be selected because applications & video conferencing require stable bandwidth.

Remote Rural Branch office

uses T1/56kbps or satellite Internet as the main connection.

use 4G LTE available

use Satellite Internet if mobile connection is poor.

use a dual-WAN router for backup connection

Install a Wi-Fi router for office users.

use a VPN to securely connect with head quarters

use a firewall to block unauthorized traffic.

use WPA3 for secure wireless access

4G LTE provides good speed at low cost.

satellite provides connectivity in remote areas.

Backup connectivity improve reliability.

The network can be upgraded to fibre in the future.

This design balance cost, reliability, security & scalability.

Wireless network Design for a university with

Introduction

A university with more than 3000 students requires a wireless network capable of supporting a large number of simultaneous users in classrooms, laboratories, & hostels. Since the number of wireless access points is limited by the budget the network should use careful access-points.

Access point placement

Access points should be placed mainly in classrooms, laboratories, & hostels where the number of users is high. Proper placement helps to provide maximum coverage with few access points.

frequency selection

The 5GHz band should be preferred because it provides higher speed & less interference. The 2.4GHz can be used where higher coverage is required. WiFi 6 can improve performance for many users.

Authentication:-

WPA2 - Enterprise or WPA3 - Enterprise should be used for secure 802.1x & Radius authentication. Separate VLANs can be created for students, staff, guests & IoT devices.

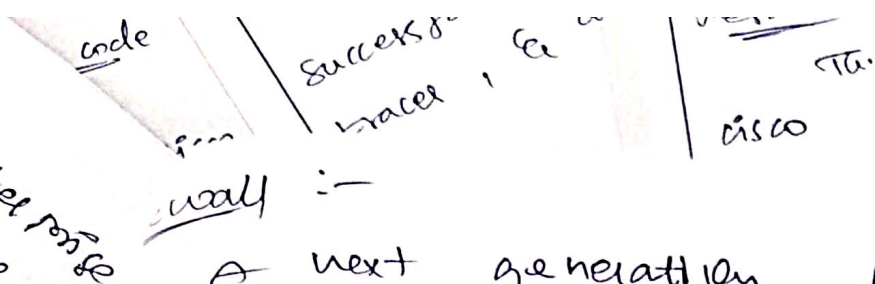
3. Financial Institution:-

Introduction:-

A financial institution handles highly sensitive customers & financial information, so its network must provide strong security while maintaining good performance & reasonable operational costs. The recommended architecture should use network segmentation, firewalls, strong authentication, intrusion detection, encryption & continuous monitoring.

Network Segmentation:-

The network should be divided into separate VLANs for employees, servers, finance systems, guests & management devices. Sensitive databases should be isolated from normal users.



Wall :-
A next generation firewall should be placed b/w the Internet & internal network. A DMZ can be used for public services such as web security.

Authentication :-
multi factor authentication, 802.1X, role based access control & VPN encryption should be used to protect sensitive customer information & prevent unauthorized access.

Intrusion Detection :-
An IDS/IPS should monitor traffic, detect suspicious activities & block threats. Logging can be used to monitor security events.

Justification :-
This architecture improves cybersecurity, protects sensitive financial data, supports compliance requirements, maintains network performance & controls operational costs.